RAJALAKSHMI ENGINEERING COLLEGE

Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:24 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:25 AM
Time taken	53 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the duplicate number in a read-only array
int findDuplicate(int* arr, int n) {
    if (n <= 1) return -1;

    // Phase 1: Finding the intersection point in the cycle
    int tortoise = arr[0];
    int hare = arr[0];

    do {
        tortoise = arr[tortoise];       // Moves 1 step
        hare = arr[arr[hare]];          // Moves 2 steps
    } while (tortoise != hare);

    // Phase 2: Finding the entrance (duplicate element) of the cycle
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];       // Both move 1 step
        hare = arr[hare];
    }

    return tortoise;
}

int main() {
    int n;

    // Read the total number of elements
    if (scanf("%d", &n) != 1) return 0;

    int* arr = (int*)malloc(n * sizeof(int));
    if (arr == NULL) return 1;

    // Read the elements line by line
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 1;
        }
    }

    // Find and print the duplicate element
    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    // Clean up memory
    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:25 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:26 AM
Time taken	42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the duplicate number using cycle detection
int findDuplicate(int* arr, int n) {
    // Phase 1: Find the intersection point of the two runners
    int tortoise = arr[0];
    int hare = arr[0];

    do {
        tortoise = arr[tortoise];       // Moves 1 step
        hare = arr[arr[hare]];          // Moves 2 steps
    } while (tortoise != hare);

    // Phase 2: Find the entrance to the cycle (the duplicate number)
    tortoise = arr[0];
    while (tortoise != hare) {
        tortoise = arr[tortoise];       // Moves 1 step
        hare = arr[hare];              // Moves 1 step
    }

    return tortoise;
}

int main() {
    int n;
    // Read the number of elements
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    // Allocate memory for the array
    int* arr = (int*)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the n elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Find and print the duplicate
    int duplicate = findDuplicate(arr, n);
    printf("%d\n", duplicate);

    // Free allocated memory
    free(arr);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:26 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:28 AM
Time taken	1 min 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void findIntersection(int arr1[], int n1, int arr2[], int n2) {
    int i = 0, j = 0;
    int first = 1; // Used to handle space formatting for the output

    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr2[j] < arr1[i]) {
            j++;
        } else {
            // Both elements are equal
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;
            i++;
            j++;
        }
    }
```

```
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        int n1;
        scanf("%d", &n1);
        int arr1[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &arr1[i]);
        }

        int n2;
        scanf("%d", &n2);
        int arr2[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &arr2[i]);
        }

        findIntersection(arr1, n1, arr2, n2);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:28 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:29 AM
Time taken	43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

void findIntersection() {
    int n1, n2;
    scanf("%d", &n1);
    int arr1[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    scanf("%d", &n2);
    int arr2[n2];
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1; // Used to handle space formatting

    while (i < n1 && j < n2) {
        if (arr1[i] < arr2[j]) {
            i++;
```

```
        } else if (arr1[i] > arr2[j]) {
            j++;
        } else {
            // Both elements are equal
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;
    while (t--) {
        findIntersection();
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:29 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:30 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int A[], int n, int k) {
    int i = 0;
    int j = 1;

    while (i < n && j < n) {
        int diff = A[j] - A[i];

        if (i != j && diff == k) {
            return 1; // Pair found
        } else if (diff < k) {
            j++;
        } else {
            i++;
            if (i == j) {
                j++;
            }
        }
    }
    return 0; // No pair found
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int A[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    // Handle negative k inputs if absolute difference isn't required,
    // but the problem states k is non-negative.
    if (k < 0) k = -k;

    int result = findPair(A, n, k);
```

```
40      int result = findPair(A, n, k);
41      printf("%d\n", result);
42
43      return 0;
44  }
45
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Tanisha S 251801350-2025-AIDS

T2

Started on	Thursday, 24 September 2026, 10:30 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:31 AM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int A[], int n, int k) {
    int i = 0, j = 1;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        int diff = A[j] - A[i];

        if (diff == k) {
            return 1;
        } else if (diff < k) {
            j++;
        } else {
            i++;
            if (i == j) {
                j++;
            }
        }
    }
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int A[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    int result = findPair(A, n, k);
```

```
    int result = findPair(A, n, k);
    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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